TC ⇒

merge Sort

$T(n) = k_1 + T(\frac{n}{2}) + T(\frac{n}{2}) + n*k$

{

// BC ($k_1$)

// left call $T(\frac{n}{2})$

// Right call $T(\frac{n}{2})$

// merge $n*k$

$= k_1 + 2T(\frac{n}{2}) + n.k$

$\boxed{T(n) = 2T(\frac{n}{2}) + n.k}$

Solve using elimination method

$\Rightarrow T(n) = 2T(\frac{n}{2}) + n*k$

$T(\frac{n}{2}) = 2T(\frac{n}{4}) + \frac{n}{2}*k$

$T(\frac{n}{4}) = 2T(\frac{n}{8}) + \frac{n}{4}*k$

$T(\frac{n}{8}) = 2T(\frac{n}{16}) + \frac{n}{8}*k$

$\vdots$

$T(1) = k$

T(1) → 1 size
1 size ke array ko sort
karne mai kitna time lagta
hai ⇒ O(1) or k

$T(n) = 2T(\frac{n}{2}) + n*k$

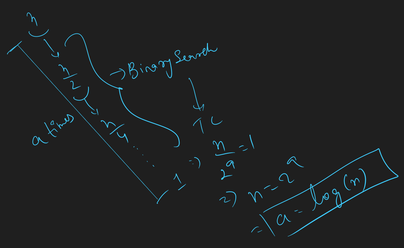
$\Rightarrow 2T(\frac{n}{2}) = 4T(\frac{n}{4}) + \frac{n}{2}*k \quad ]\times 2$

$a-1 \quad 4T(\frac{n}{4}) = 8T(\frac{n}{8}) + \frac{n}{4}*k \quad ]\times 4$

a times $\quad 8T(\frac{n}{8}) = 16T(\frac{n}{16}) + \frac{n}{8}*k \quad ]\times 8$

$= k$

$T(n) = (a-1)*n*k + k$




$T(n) = (a-1)*n*k + k$

$= \log(n-1) * n*k + k \rightarrow$ Constant k neglect

$= \log(n-1)*nk$

$= nk\log n - nk$

smaller than $n\log n$ in terms of big O.

$= n\log n$

$\Rightarrow T(n) = k_1 + (a-1)(n*k)$

$= (\log n - 1)(n*k)$

$= n\log n - n$

$= O(n\log n)$